# Degree Correlations and Network Assortativity

### Background

In complex networks, not all connections are random — the degree of a node often influences the degrees of its neighbors. This dependency is known as degree–degree correlation, and it significantly affects network robustness, spreading processes, and modularity.

## Definitions

- **Degree correlation**: The tendency of nodes to connect to others with similar (or dissimilar) degrees.
  Measured by the **correlation coefficient** *r* between the degrees at either end of an edge.
- **Assortativity**:
  r > 0 — high-degree nodes preferentially connect to other high-degree nodes.
  Common in **social networks**.
- **Disassortativity**:
  r < 0 — high-degree nodes connect to low-degree nodes.
  Common in **technological or biological networks** (e.g., Internet, protein networks).
- **Structural cutoff**:
  In finite networks, there's a limit on the maximum degree $k_{max}$ = compatible with the network's size and degree distribution.

  $$k_{cut} \sim \sqrt{\bar{k} \cdot N}$$

  Beyond this, correlations emerge purely due to structural constraints rather than true assortativity.

- **Average nearest-neighbor degree:** $k_{nn}(k)$ quantifies how the degree of a node correlates with the degrees of its neighbors. It provides a local measure of degree correlations in a network.

$$k_{nn}(k) = \frac{1}{N_k} \sum_{i|k_i=k} \frac{1}{k_i} \sum_{j \in N(i)} k_j$$

## Task 1: Computing Degree Correlations

1. Generate a Barabasi-Albert undirected network with N=1000 and m=3
2. Compute the assortativity coefficient
3. Plot the **average nearest-neighbor degree** $k_{nn}(k)$. Use log–log scales and comment on whether the network shows assortative or disassortative tendencies.

## Task 2: Manipulating Correlations

1. **Generate a correlated network** using `nx.random_degree_sequence_graph()` with degree-preserving rewiring:
    - Start from a degree sequence following a power law.
    - Apply rewiring steps to **increase** or **decrease** assortativity (you can implement a simple edge-swap rule).
2. Compute and compare:
    - Degree distribution P(k)
    - $k_{nn}(k)$
    - Assortativity coefficient r
3. **Question:**
   How does increasing assortativity affect:
    - The clustering coefficient?
    - The network's diameter?

## Task 3: Correlations in Real Networks

1. Download a real-world network dataset (e.g., social, technological, biological). Example: Zachary's Karate Club, Facebook ego-network, or a protein–protein interaction network.
2. Compute:
    - r
    - $k_{nn}(k)$
    - $k_{cut}$
3. Compare these metrics with those from a randomized version of the same network (using degree-preserving randomization).